

July 1, 2026 1:24 PM

$$\begin{array}{l}
 x_1 \\
 x_2 \\
 \vdots \\
 x_N
 \end{array}
 \quad
 \mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\mu = 700$$

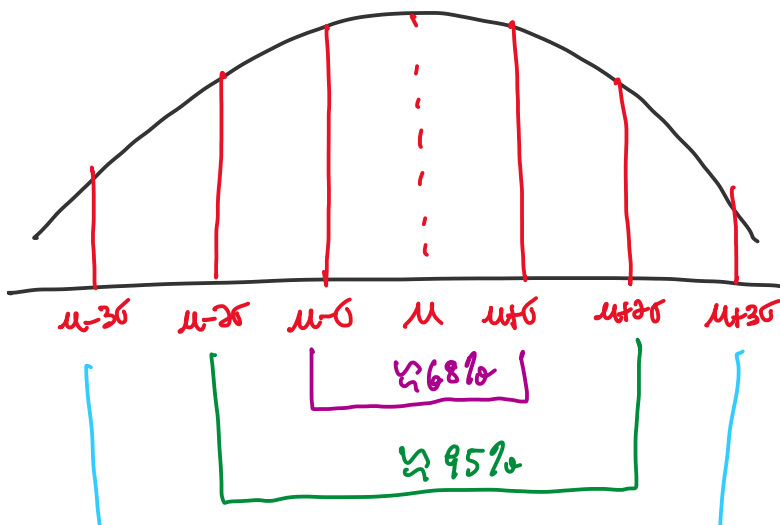
$$\begin{array}{l}
 (x_1 - \mu)^2 \\
 (x_2 - \mu)^2 \\
 (x_3 - \mu)^2 \\
 \vdots \\
 (x_N - \mu)^2
 \end{array}$$

$$\sigma^2 = \text{Variance} = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N} = \frac{\sum_{i=1}^N x_i^2 - N\mu^2}{N}$$

$$\sigma = \text{standard deviation} = \sqrt{\sigma^2}$$

$$\sigma = 30$$

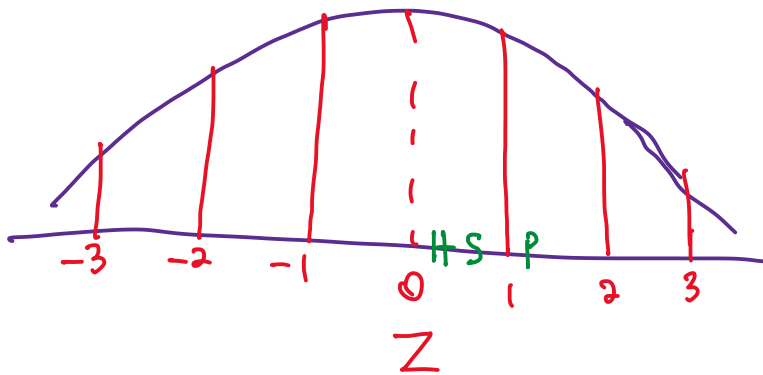
Empirical Rule



Σ 99%

$$Z = \frac{X - \mu}{\sigma}$$

Physics	History	Stats
$\mu = 70$	$\mu = 80$	$\mu = 75$
$\sigma = 5$	$\sigma = 3$	$\sigma = 4$
$X = 76$	$X = 81$	$X = 78$
$Z = \frac{76 - 70}{5} = 1.2$	$Z = \frac{81 - 80}{3} = \frac{1}{3}$	$Z = \frac{78 - 75}{4} = \frac{3}{4}$



$|Z| \geq 3$, outlier

$2 \leq |Z| < 3$, somewhat an outlier

X	N=5
7	
5	
1	
4	
2	

1. μ

2. σ^2

3. σ

4. Z values for each X.

